

BE 308: Introduction to Biomanufacturing

Course Code	BE 308
Course Name	Introduction to Biomanufacturing
L-T-P-C	3-0-2-4
Prerequisites	IC 136 Understanding Biotechnology and its Applications or Consent of Faculty Member
Intended for	UG
Mutual Exclusion	NA
Approval	50th BoA (Jul 2023)
Revised	64th BoA (May 2026) - syllabus revised

Preamble

This course introduces the principles and practices of biomanufacturing, focusing on the production of value-added products using biological systems. It covers upstream and downstream processing, bioreactor design, scale-up strategies, quality control, regulatory frameworks, and emerging trends such as synthetic biology and biofoundries. The course integrates theory with case studies from pharmaceuticals, biofuels, food, and environmental sectors.

Course Modules

- Module 1: Introduction to Biomanufacturing:** Definition, scope, and importance of biomanufacturing; Introduction to biologicals and biomolecules for Industrial Application, Stages of biomanufacturing and overview of bioprocess workflow; Types of bioproducts: pharmaceuticals, vaccines, enzymes, biofuels. Historical development and industrial relevance; Overview of biomanufactured products: Pharmaceuticals (antibiotics, vaccines); biofuels (bioethanol, biogas, biodiesel); food & enzymes; Introduction to expression systems, nutritional strategies, and cell culture media with relevant industrial case studies. (12 Hours)
- Module 2: Production, Isolation & Purification:** Media preparation, sterilization, inoculum development, and fermentation strategies (batch, fed-batch, continuous). Bioreactor design, operation, process monitoring, aeration, agitation, oxygen transfer, and scale-up considerations. Cell harvesting, centrifugation, filtration, chromatography, membrane separation, ultrafiltration, diafiltration, viral clearance, and product purification. (12 Hours)
- Module 3: Industrial Scaleup and Regulatory Framework:** Process scale-up, process optimization, techno-economic analysis (TEA), and life cycle assessment (LCA). Process validation and characterization. Good Manufacturing Practices (GMP), Quality Assurance (QA), Quality Control (QC), documentation, and regulatory guidelines (CDSCO, US FDA, EMA). Biosafety, ethical considerations, and product approval processes. (12 Hours)
- Module 4: Industrial Applications and Emerging Trends:** Biomanufacturing applications in tissue engineering, monoclonal antibodies, recombinant therapeutics, gene therapy, and mRNA vaccines. AI in bioprocess optimisation, sustainable biomanufacturing, and industrial case studies. (06 Hours)

Laboratory Modules (28 hours)

1. Media preparation and sterilisation
2. Cell separation
3. Chromatography
4. Ultrafiltration and diafiltration
5. Biomass estimation and product analysis
6. Alginate bead encapsulation

Text Books

1. Shuler & Kargi – *Bioprocess Engineering: Basic Concepts*
2. Nielsen & Villadsen – *Bioreaction Engineering Principles*

References

1. Gilleskie, Gary, Charles Rutter, and Becky McCuen, *Biopharmaceutical Manufacturing: Principles, Processes, and Practices*, Walter de Gruyter GmbH & Co KG, 2021.
2. Doran – *Bioprocess Engineering Principles*
3. Atala, Anthony, and Julie Allickson, eds., *Translational regenerative medicine*, Academic Press, 2014.
4. Wei Wang, Manmohan Singh, *Biological Drug Products: Development and Strategies*, Wiley, 2013.
5. Stanbury et al. – *Principles of Fermentation Technology*
6. Harrison et al. – *Bioseparations Science and Engineering*
7. Ratledge & Kristiansen – *Basic Biotechnology*